

SOP: Sulfur Extraction

NiMBLE Project

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Reagents:

- Carbon Disulfide ([SDS](#))

Glassware & Equipment:

- Aluminum cups (WB353)
- P4 Grade Filter Paper (WB353)
- Erlenmeyer/Conical flasks (WB334) (~50–70 mL)
- Solids Funnel (WB334)
- Filter Funnel (WB334)
- Filter Flask/Vacuum Flask (WB334)
- Büchner flask (WB334)
- PTFE (Teflon) Beaker (WB334)

PPE:

- Nitrile Gloves
- Lab Coat
- Safety Glasses
- Fumehood

Safety

The primary concern in this procedure is carbon disulfide (CS_2). Please read the SDS before proceeding. CS_2 is highly toxic, flammable, and vaporizes very easily.

- ⚠ Must be used in a **functioning** fumehood.
- ⚠ Avoid any sources of ignition.
- ⚠ Nitrile gloves are permeable to CS_2 .
- ⚠ Double glove to minimize exposure.
- ⚠ Remove the gloves if a large quantity spills on them.
- ⚠ Lower the fumehood sash as much as possible.

Pre-procedure

1. Make sure the weighing scale in WB334 is zeroed and the level bubble is correctly centered.
2. Ensure all the samples are dry. Carbon disulfide is not miscible with water and water will severely limit sulfur extraction.
3. CS_2 has high surface tension, and an affinity for glass. Transfer an aliquot of CS_2 to the PTFE beaker and pour from this instead.

4. When not in use, close the CS₂ bottle to minimize evaporative losses.

Procedure

1. Place an Erlenmeyer flask onto the scale and tare.
2. Place a funnel on top of the flask.
3. Add your solids to the flask — ~1–3 g of solids.
4. Remove the funnel and weigh; record the mass added.
5. If the solid is clumped, use a spatula to break it up. Re-weigh the sample if any sample was unintentionally removed during crushing.
6. Move the sample to the fumehood and add CS₂ to the sample. Ensure the solids are submerged¹.
7. Seal the flask with aluminum foil and gently stir.
8. Leave to rest for 1 hour or more.
9. While the flask rests, take an aluminum cup per sample and weigh it. Record its weight and label it appropriately.
10. After the flask has rested, assemble the vacuum filtration equipment. The filter funnel fits into the mouth of the filter/vacuum flask.
11. Attach the vacuum pipe to the flask and turn on the vacuum.
12. Place a leaf of filter paper on top of the funnel. The vacuum should hold the paper in place.
13. Place the Büchner flask on top of the funnel to create a watertight seal and secure with the clamp.
14. Check the assembly to ensure everything is secure and stable.
15. Remove the foil from the sample flask and gently swirl the glass to suspend the solids.
16. Pour the mixture into the Büchner flask. The liquid should pass through unimpeded and the solids should be retained back.
17. Rinse the Erlenmeyer flask twice with a little CS₂, pouring it back into the flask each time to minimize losses.
18. Set the flask aside inside the fumehood.
19. Carefully shut off the vacuum as you pull the pipe off the flask. Turning off the vacuum before removing the pipe can cause water to enter the vessel, and thus, this should be avoided.
20. Disassemble the clamp, Büchner flask, and funnel.
21. Carefully pour the filtrate from the filter flask into the appropriate aluminum cup. Remember that pouring too fast can lead to splashes and spills, and pouring too slowly can lead to the CS₂ streaming down the side of the glass flask.
22. Similar to the Erlenmeyer flask, add small amounts of CS₂ and rinse the filter flask. Pour this CS₂ into the cup too.

¹Sulfur has a solubility of 0.45 g/mL in CS₂, and therefore, for 1 g of sulfur in a sample, only ~2.2 mL of CS₂ is needed. Use ~10 mL of CS₂.

23. Leave the cup inside the fumehood for the CS_2 to dry. This can take anywhere from 3 to 6 hours.
24. The evaporation of the CS_2 can cause cold water to condense onto the cup. If this happens, the best option is to also wait for the water to evaporate. If that is not possible, wait for all the CS_2 to evaporate and then place the cup in the 40°C oven for the water to dry. **Remember, CS_2 is extremely flammable, and thus this step should only be done if absolutely *necessary* and with the utmost care.**
25. Finally, weigh the dry aluminum cup. If all the steps have been executed correctly and there was sulfur present in the sample, there will be yellow crystals in the cup and it should weigh more. Record the total weight and subtract from the cup weight to compute the sulfur mass.
26. Enter the data on the Solid Sulfur sheet on the NiMBLE data tracking excel with the associated sample number.